

STAT 100 - Important Code for Week 5 (Simulation-Based Confidence Intervals)

Functions

Set **seed** to save randomly-generated information over time (put this at top of r-chunk)

- `set.seed(---)`
 - `set.seed(1)`

Compute the **sample statistic**, which will be center of **confidence interval**

- `SAMPLE-STATISTIC <- DATASET %>%`
 `drop_na(RESPONSE-VAR) %>%`
 `specify(response = RESPONSE-VAR) %>%`
 `calculate(stat = "TYPE-OF-STAT")`
- `x_bar <- movies %>%`
 `drop_na(OpeningWeekend) %>%`
 `specify(response = OpeningWeekend) %>%`
 `calculate(stat = "mean")`

Generate and graph a **bootstrap distribution** from a **mean**

- `BOOTSTRAP-DIST <- DATASET %>%`
 `drop_na(RESPONSE-VAR) %>%`
 `specify(response = RESPONSE-VAR) %>%`
 `generate(reps = 1000, type = "bootstrap") %>%`
 `calculate(stat = "TYPE-OF-STAT")`
 `ggplot(data = BOOTSTRAP-DIST,`
 `mapping = aes(x = stat)) +`
 `geom_histogram()`
- `bootstrap_dist <- movies %>%`
 `drop_na(OpeningWeekend) %>%`
 `specify(response = OpeningWeekend) %>%`
 `generate(reps = 1000, type = "bootstrap") %>%`
 `calculate(stat = "mean")`
 `ggplot(data = bootstrap_dist,`
 `mapping = aes(x = stat)) +`
 `geom_histogram()`

Generate and graph a **bootstrap distribution** from a **difference in means**

- `BOOTSTRAP-DIST <- DATASET %>%`
 `drop_na(RESPONSE-VAR) %>%`
 `specify(RESPONSE-VAR ~ CAT-EXPLANATORY-VAR) %>%`

```

generate(reps = 1000, type = "bootstrap") %>%
calculate(stat = "diff in means",
order = c("CATEGORY-1", "CATEGORY-2"))
ggplot(data = BOOTSTRAP-DIST,
mapping = aes(x = stat)) +
geom_histogram()
- bootstrap_dist <- movies %>%
  drop_na(OpeningWeekend) %>%
  filter(Genre %in% c("Action", "Drama")) %>%
  specify(OpeningWeekend ~ Genre) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in means",
order = c("Action", "Drama"))
ggplot(data = bootstrap_dist,
mapping = aes(x = stat)) +
geom_histogram()

```

Compute **confidence interval via SE method**

```

- CONF-INTERVAL <- BOOTSTRAP-DIST %>%
  get_confidence_interval(type = "se", level = CONF-LEVEL,
point_estimate = SAMPLE-STATISTIC)
- ci <- bootstrap_dist %>%
  get_confidence_interval(type = "se", level = 0.95,
point_estimate = x_bar)

```

Compute **confidence interval via percentile method**

```

- CONF-INTERVAL <- BOOTSTRAP-DIST %>%
  get_confidence_interval(type = "percentile", level = CONF-LEVEL)
- ci <- bootstrap_dist %>%
  get_confidence_interval(type = "percentile", level = 0.95)

```

Quickly graph **bootstrap distribution**

```

- BOOTSTRAP-DIST %>%
  visualize()
- bootstrap_dist %>%
  visualize()

```